Annapurna Capital Technologies — TAM / SAM / SOM Independent Rebuild

Task 4 Market-Sizing Note

Given figures (Case Study, Section 3): TAM ₹8,40,000 Cr · SAM ₹2,40,000 Cr (28.6% of TAM) · SOM ₹6,000 Cr/yr (longer-term aspiration)

1. TAM — Total Addressable Market

1a. Bottom-Up Build

Using FY26 actual production data (Ministry of Agriculture, 3rd Advance Estimates):

Crop Category	Production (Million Tonnes)	Blended Price (₹/tonne, MSP-based)	Value (₹ Cr)
Cereals (rice, wheat, coarse grains)	347.6	21,000	7,29,882
Pulses	29.0	65,000	1,88,500
Oilseeds	43.1	55,000	2,36,824
Gross Storable Crop Value			**11,55,207**

Correction applied — marketable surplus: not all production is ever sold; a meaningful share is retained by farmers for self-consumption, seed, and animal feed, and can never be warehouse-financed. Applying a blended 68% marketable-surplus factor (lower for staple foodgrains kept for household use, higher for cash-oriented pulses/oilseeds):

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Adjusted Bottom-Up TAM = ₹11,55,207 Cr × 68% = ₹7,85,541 Cr
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1b. Top-Down Cross-Check

India's Agriculture GVA (FY26) = ₹52,08,800 Cr. Crops represent ~54% of this (~₹28,13,000 Cr); cereals/pulses/oilseeds represent ~45% of crop GVA (excluding horticulture, cotton, sugarcane, plantation crops) → ~₹12,66,000 Cr gross, adjusted by the same 68% marketable-surplus factor:

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Top-Down TAM = ₹8,60,000 Cr
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1c. Reconciliation

Both independently-derived methods converge tightly (₹7.86–8.60 lakh Cr) and closely validate the case's stated ₹8,40,000 Cr — but through a materially different, defensible path. **A naive gross-production estimate (₹11.55 lakh Cr) would have overstated TAM by ~38%**; the marketable-surplus correction is the key insight that reconciles bottom-up reality with the case's top-down number.

Independent TAM: ₹8,00,000 – 8,60,000 Cr (midpoint ≈ ₹8,23,000 Cr, consistent with the case)

2. SAM — Serviceable Addressable Market

The case defines SAM by bank/NBFC partner geographic coverage, arriving at 28.6% of TAM (₹2,40,000 Cr). Our independent rebuild uses the same logic but with current business facts: Annapurna is targeting **16+ states** post-Series B (up from 8 today). India's top agricultural-producing states — UP, Punjab, Haryana, MP, Maharashtra, Rajasthan, Karnataka, AP, Telangana, Bihar, West Bengal, Gujarat, Chhattisgarh, Odisha, Tamil Nadu, and Assam — collectively account for **~80% of national storable-crop marketable surplus**.

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Independent SAM = 80% × ₹8,40,000 Cr = ₹6,72,000 Cr
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This is ~2.8x the case's stated SAM. The gap exists because the case's 28.6% figure appears to still reflect a warehouse-density-capped assumption, whereas the case itself states that owning mini-warehouses removes that exact cap. If that premise is taken at face value, SAM should scale with bank-partner geographic reach (80% of national output), not pre-existing warehouse density (28.6%).

3. SOM — Serviceable Obtainable Market

Rather than a separate top-down guess, SOM is taken directly from the Task 1 driver-based forecast already built for this assignment:

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FY31E Annual Loan Disbursement Volume = ₹1,979 Cr  ≈ 0.29% of independent SAM (₹6,72,000 Cr)
Case's long-term SOM aspiration      = ₹6,000 Cr/yr ≈ 0.89% of independent SAM
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The ₹6,000 Cr figure should be read as a post-FY31 aspiration — further share-of-SAM capture beyond this model's 5-year explicit window — not a target this forecast is required to hit.

4. Summary Table

Metric	Case's Stated Figure	Independent Rebuild	Basis
TAM	₹8,40,000 Cr	₹8,00,000–8,60,000 Cr	Converges with case; bottom-up validated via marketable-surplus-adjusted FY26 production data, cross-checked top-down via Agri GVA
SAM	₹2,40,000 Cr (28.6%)	₹6,72,000 Cr (~80%)	Diverges materially; based on 16-state geographic bank-partner coverage rather than legacy warehouse-density assumption
SOM	₹6,000 Cr/yr (aspiration)	₹1,979 Cr/yr (FY31E)	Tied directly to Task 1's own forecast, not a separate estimate

5. Critique — Does Owning Mini-Warehouses Genuinely Remove the "Warehouse Density" Constraint on SAM?

Partially — it removes one real constraint but replaces it with a different one, rather than eliminating the constraint altogether.

What is genuinely removed: Pre-existing physical warehouse density. Previously, if a district had no trustworthy warehouse, Annapurna simply could not operate there, regardless of whether a bank partner was willing to lend. Owning mini-warehouses lets Annapurna build storage anywhere a bank/NBFC partner operates — this is real, and it's exactly why the independent SAM (₹6,72,000 Cr, geography-based) comes out substantially higher than the case's warehouse-density-implied figure.

What is not removed — only replaced: Capital and execution pace. By FY31E, the model plans for only 290 owned mini-warehouses at ~700 tonnes average capacity — roughly **2.03 lakh tonnes of owned storage nationally**, a rounding error against hundreds of millions of tonnes of addressable marketable surplus. Partner warehouse growth (700 planned by FY31) is itself gated by finding compliant operators willing to install IoT sensors and meet WDRA standards. And even with unlimited warehouse capacity, the **11 partner banks/NBFCs' own balance sheet capacity and risk appetite** cap how much lending volume can actually flow through the system.

Conclusion: Owning mini-warehouses converts SAM from a *fixed, binary, geography-capped* number into a genuinely larger addressable base — but replaces that cap with a *scalable-but-still-real* constraint (capital deployment speed + partner lending capacity), rather than removing the constraint altogether. This is a materially better constraint to have, since it scales with capital rather than being permanently capped by geography — but it is not the same as an unconstrained market, and the Series B use-of-funds allocation (55% to warehouse network expansion) should be read as directly funding this capital constraint, not a marketing-driven growth spend.